

# STRUCTURAL ANALYSIS AND DESIGN REPORT



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## Table of Contents

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| <b>SECTION A – STRUCTURAL ANALYSIS AND DESIGN .....</b>          | <b>3</b>  |
| <b>A1. Determination of Actions and Load Paths .....</b>         | <b>3</b>  |
| <b>A2. Steel Beam Design for Removed Load-Bearing Wall .....</b> | <b>5</b>  |
| Design of masonry Bearing at each end .....                      | 7         |
| Concrete padstone design .....                                   | 8         |
| <b>A3. Reinforced Concrete Slab Design Calculation .....</b>     | <b>9</b>  |
| <b>A4. Timber Floor Joist Design (EN 1995) .....</b>             | <b>13</b> |
| <b>A5. Masonry Wall Panel Check.....</b>                         | <b>15</b> |
| <b>SECTION B – FRAME ANALYSIS: MOMENT DISTRIBUTION .....</b>     | <b>18</b> |
| Stiffness Factor calculation .....                               | 19        |
| Distribution Factor Calculations .....                           | 19        |
| Moment Distribution Calculation .....                            | 20        |
| <b>SECTION C – SPACE GASS FRAME ANALYSIS .....</b>               | <b>20</b> |
| C1. Loading Condition.....                                       | 20        |
| C2. Bending Moment Diagram .....                                 | 21        |
| C3. Shear Force Diagram .....                                    | 21        |

## SECTION A – STRUCTURAL ANALYSIS AND DESIGN

### A1. Determination of Actions and Load Paths

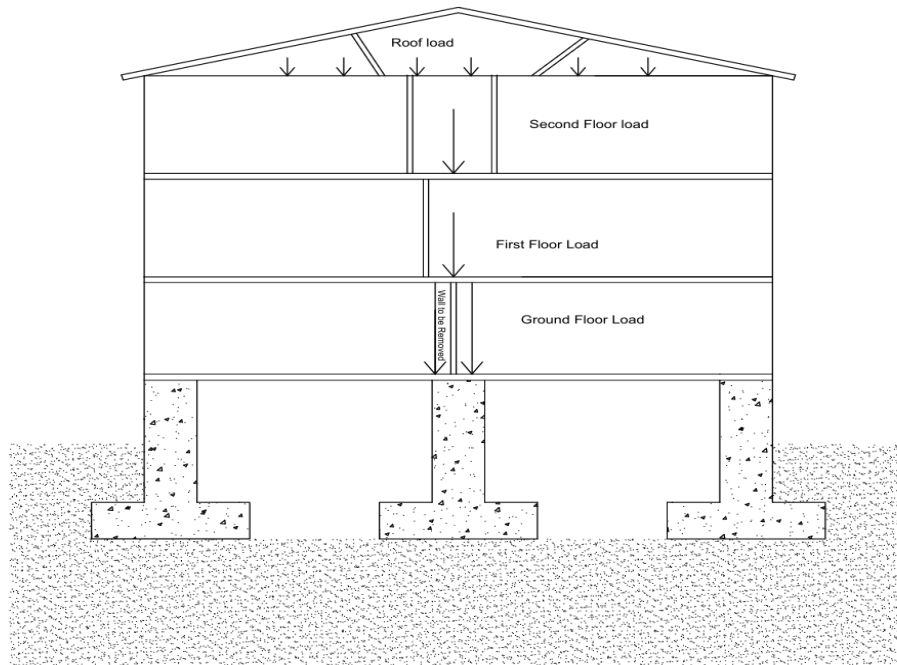


Fig: Load path

The loads that are gathered by the rafters or trusses are conveyed to wall plates or supporting beams. Wall plates serve to distribute the load uniformly and avert localized overstressing of the supporting masonry. In instances where beams are utilized, they bear the loads to load-bearing walls or columns at specific points. From the wall plates or beams, the loads are directed vertically downward into the load-bearing walls or columns. Masonry walls primarily support load through compression and transmit it continuously down the height of the structure. As the load descends, it naturally disperses through the masonry, thereby diminishing the stress at lower levels.

At the ground level, the accumulated loads from the walls or columns are transferred into the foundations. The foundations are engineered to distribute the load over a broader area, preventing the pressure exerted on the soil from surpassing its safe bearing capacity. This design mitigates the risk of excessive settlement or ground failure. Ultimately, the load is conveyed from the foundations into the underlying soil, which supports the structure. This process completes the continuous load path from the roof to the ground, guaranteeing that all applied loads are effectively resisted and transferred without overstressing any component of the structure.

| References                                                                                    | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcomes                                                              |
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| <p>Cobb's Fionna Book<br/>Page:91<br/>Page:91<br/>page:91</p> <p>Cobb's Fionna book pg.93</p> | <p>First floor Load (Dead + Imposed)<br/>Dead Load (<math>G_k</math>):<br/>Floor construction (typical UK timber floor):</p> <p>Timber joists + floorboards: <math>0.5 \text{ kN/m}^2</math><br/>Ceiling + services: <math>0.3 \text{ kN/m}^2</math><br/>Partitions allowance: <math>0.5 \text{ kN/m}^2</math><br/>Total dead load= <math>1.3 \text{ kN/m}^2</math><br/>Tributary width: Looking at the plan, the wall supports floor area on both sides.</p> <p>Estimate tributary width:</p> <p>Kitchen/Utility side =1.5m<br/>Dining Room side= 1.5m<br/>Total tributary width= 3.0m<br/>Dead load per meter of wall( <math>G_k</math>) = <math>1.3 \text{ kN/m}^2 \times 3.0\text{m}</math><br/>= <math>3.9 \text{ kN/m}</math></p> <p>Wall load (First Floor Wall)<br/>Load from first floor to existing wall = <math>0.1 \times 20 \times 2.40</math><br/>= <math>4.80 \text{ kN/m}^2</math><br/>Load for supported wall in 3m span = <math>4.80 \times 3</math><br/>= <math>14.4 \text{ kN/m}</math></p> <p>Live load (<math>Q_k</math>) = <math>1.5 \text{ kN/m}^2</math><br/>Load over 3m span = <math>1.5 \times 3</math><br/>= <math>4.5 \text{ kN/m}</math></p> <p>Total Characteristic load= <math>G_k + Q_k</math><br/>= <math>18.3 + 4.5</math><br/>= <math>22.8 \text{ kN/m}</math></p> | <p>Total characteristic load<br/>= <math>22.8 \text{ kN/m}</math></p> |

| References                                          | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcomes                                                                                                       |
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| <p>see EN 771-3</p> <p>Cobb's Fionna book pg.93</p> | <p><b>A2. Steel Beam Design for Removed Load-Bearing Wall</b></p> <p>Span of a wall = 3m<br/> Thickness of a wall = 100mm<br/> Height of a wall = 2.40m<br/> Now,<br/> Dead load per meter of wall( <math>G_k</math>) = <math>1.3 \text{ kN/m}^2 \times 3.0\text{m}</math><br/> = 3.9 kN/m</p> <p>Load from first floor to existing wall = <math>0.1 \times 20 \times 2.40</math><br/> = 4.80 kN/m<sup>2</sup></p> <p>Load over 3m span of a wall ( <math>Q_k</math>) = <math>4.80 \times 3</math><br/> = 14.4 kN/m</p> <p>Live load = <math>1.5 \text{ kN/m}^2</math><br/> Load over 3m span = <math>1.5 \times 3</math><br/> = 4.5 kN/m</p> <p>Total load on existing load bearing wall = <math>G_k + Q_k</math><br/> = <math>18.3 + 4.5 = 22.8 \text{ kN/m}</math></p> <p>Beam design and calculation<br/> Ultimate load combination (ULS)<br/> <math>F_{ed} = 1.35 \times 18.3 + 1.5 \times 4.5</math><br/> = 31.45 kN/m</p> <p>Maximum Bending Moment</p> $M_{ed} = \frac{wl^2}{8}$ $= \frac{31.45 \times 3^2}{8} = 35.38 \text{ kNm}$ <p>Minimum required plastic moment of resistance about major axis (y-y), Wply is given by:</p> $W_{ply} \geq \frac{M_{pl} R_d}{f_y} = \frac{M_{ed}}{f_y} = \frac{27.68 \times 10^6}{275} = 100.65 \times 10^3 \text{ mm}^3$ <p>See Steel design data Table B.1 pg. B-7</p> <p>Depth of a section = 117.4 mm<br/> Width of a section = 101.2 mm<br/> Web thickness = 4.8 mm<br/> Flange thickness = 7.9 mm<br/> Root Radius = 7.6 mm<br/> Area = 24.3 CM<sup>2</sup><br/> Try 152 × 89 × 16                      Wply = 171 CM<sup>2</sup></p> <p>Checking the classification of cross-section</p> $\epsilon = \sqrt{\frac{235}{f_y}} = \sqrt{\frac{235}{275}} = 0.92$ | <p>Total load = 22.8 kN/m</p> <p><math>F_{ed} = 31.45 \text{ kN/m}</math></p> <p>Wply = 171 CM<sup>2</sup></p> |

| References                                                                | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Outcomes                                                                                          |
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| <p>Table 14.1<br/>BS EN 1993-1-1</p> <p>NA to BS EN<br/>1993-1-1:2005</p> | $\frac{C}{t_f} = \frac{(b - t_w - 2r)/2}{t_f} = \frac{(101.2 - 4.8 - 2 \times 7.6)/2}{7.9} = 5.1 \leq 9\epsilon = 8.28 \text{ (class 1)}$ $\frac{d}{t_f} = \frac{146.8}{7.9} = 18.60 \leq 72\epsilon$ $= 27.06 \leq 72 \times 0.92$ $= 27.06 \leq 66.24 \quad (\text{Class 1})$ <p>As assume <b>178 × 102 × 19</b> belongs to class 1</p> <p>Resistance of cross section to Bending Moment</p> $M_{plRD} = W_{ply} \times f_y$ $= 171 \times 10^3 \times 275$ $= 47.02 \text{ kNm}$ $\frac{M_{ed}}{M_{plRd}} = \frac{35.38}{47.02} \leq 1 = 0.7 \text{ (ok)}$ <p>Resistance of cross-section to shear</p> $V_{PLRD} = \frac{A_v \times f_y}{\sqrt{3}} = \frac{[A - 2bt_f + (t_w + 2r)t_f] \times f_y}{\sqrt{3}}$ $=$ $\frac{[24.3 \times 10^2 - 2 \times 89 \times 7.9 + (4.8 + 2 \times 7.6) \times 7.9] \times 275}{\sqrt{3}}$ $= 187.7 \text{ kN}$ <p>Now,</p> $\frac{V_{ed}}{V_{PLRD}} = \frac{F_{ed}/2}{128.7} = \frac{31.45 \times 3/2}{187.7} = 0.2 \leq 1 \text{ (ok)}$ <p><b>Vertical deflection at serviceability</b></p> <p><i>Allowable Deflection</i></p> $\delta_{allowable} = \frac{span}{360} = \frac{3000}{360} = 8.3 \text{ mm}$ <p><i>Actual Deflection</i></p> $\delta_{max} = \frac{5wl^3}{384EI}$ $= \frac{5 \times 18.24 \times 3^3}{384 \times 2.1 \times 10^8 \times 8.34 \times 10^{-6}} = 3.6 \text{ mm}$ <p><b>Serviceability check</b></p> $\frac{\delta_{max}}{\delta_{allowable}} = \frac{3.6}{8.33} = 0.41 < 1 \text{ (ok)}$ | <p><math>V_{PLRD} = 187.7 \text{ kN}</math></p> <p><math>\delta_{max} = 3.6 \text{ mm}</math></p> |

| References                                  | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcomes                                                                                                                            |
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| EN 771-3:2011<br>EN 1996-1-1:2005 Table 2.3 | <p align="center"><b>Design of masonry Bearing at each end</b></p> <p>Chosen beam section = 177 × 89 × 16 UB</p> <p>Span = 3m</p> <p><math>F_{ed} = 31.45 \text{ Kn/m}</math></p> <p><math>V_{ed} = 47.1 \text{ kn}</math></p> <p>1. Required bearing length</p> <p>Assume block masonry (<math>f_k</math>) = 10 N/mm<sup>2</sup></p> <p>Partial factor (<math>\gamma_m</math>) = 2.5</p> <p>Now,</p> <p>Design bearing strength(<math>f_d</math>) = <math>\frac{f_k}{\gamma_m} = \frac{10}{2.5} = 4 \text{ N/mm}^2</math></p> <p>Bearing area required(<math>A_{req}</math>) = <math>\frac{V_{ed}}{f_d} = \frac{47.1 \times 10^3}{4} = 11775 \text{ mm}^2</math></p> <p>With beam flange width (b) = 89 mm</p> <p>Required bearing length(<math>L_{req}</math>) = <math>A_{req} / b</math></p> <p><math>= \frac{11775}{89} = 132.65 \text{ mm}</math></p> <p>Per BS 5950 or Eurocode 3:</p> <p>Since <math>L_{req} = 132.65 \text{ mm} &gt; 100 \text{ mm}</math>, we need more than minimum.</p> <p>Minimum bearing length (<math>b_1</math>) = 135 mm (to prevent web crushing and ensure stability)</p> <p>Check Web Crushing/Crippling</p> <p>For stiff bearing (with padstone), checking local resistance:</p> <p>Bearing resistance(<math>f_{wRd} = \frac{t_w \times (b_1 + n_1) \times f_y}{\gamma_m}</math>)</p> <p>Where,</p> <p><math>t_w = 4.8 \text{ mm}</math></p> <p><math>b_1 = 135 \text{ mm}</math></p> <p><math>n_1 = 2\sqrt{2}(7.9 + 7.6) = 43.8 \text{ mm}</math></p> <p><math>f_y = 275 \text{ N/mm}^2</math></p> <p><math>\gamma_{mo} = 1.0</math></p> <p><math>F_{wRd} = \frac{4.8 \times (135 + 43.8) \times 275}{1} = 236.01 \text{ Kn}</math></p> <p>Checking,</p> <p><math>\frac{V_{ed}}{F_{wRd}} = \frac{47.1}{236.01} = 0.1 &gt; 1 \text{ (ok)}</math></p> | <p><math>f_d = 4 \text{ N/mm}^2</math></p> <p><math>b_1 = 135 \text{ mm}</math></p> <p><math>f_{wRd} = 236.01 \text{ Kn}</math></p> |



| References                                                                                                    | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outcomes        |
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| <p>Cobb's Fionna book<br/>Pg. 122</p> <p>BS EN 1992-1:2004<br/>EN 1992-1-1:200(E)<br/>Table 4.4N and 4.5N</p> | <p><b>A3. Reinforced Concrete Slab Design Calculation</b></p> <p>Width (b)= 1000mm<br/> Characteristic strength of concrete (<math>f_{ck}</math>) = 37N/mm<sup>2</sup><br/> Characteristic strength of steel (<math>f_y</math>) = 500N/mm<sup>2</sup><br/> Longer side (<math>l_y</math>)=3m<br/> Shorter side (<math>l_x</math>) =1.5m<br/> Now,<br/> Checking for the type of slab,</p> $\frac{l_y}{l_x} \geq 2 = \frac{3}{1.5} = 2$ <p>Hence, it is greater than 2 so it is a one-way slab.</p> <p>Estimating thickness/depth considering deflection control:<br/> Overall depth = <math>\frac{L}{22} = \frac{2930}{22} = 136.18 \approx 140\text{mm}</math></p> <p>Minimum cover = 30 mm (considering xc1 class S3 structural class)</p> <p>Now,<br/> live load = 1.5kN/m<sup>2</sup><br/> RC Density = 25 kN/m<sup>3</sup></p> <p>ULS partial factor (<math>Y_g = 1.35</math>      <math>Y_q = 1.50</math>)<br/> Now,<br/> Self-weight of slab = <math>140 \times 25 \times 10^{-3}</math><br/> <math>= 3.5 \text{ kN/m}^2</math><br/> Floor finish = 1 kN/m<sup>2</sup><br/> Total permanent load = 3.5+1<br/> <math>= 4.5\text{kN/m}^2</math></p> <p>For xc1 assume cover is 30mm allowing say 3mm as a half diameter of a reinforcement.<br/> Effective depth = 140-30-3<br/> <math>= 107\text{mm}</math></p> <p>For 1m width of a slab,<br/> Ultimate load (<math>V_{ed}</math>) = <math>1.35g_k + 1.50q_k</math><br/> <math>= 1.35 \times 4.5 + 1.5 \times 1.5</math><br/> <math>= 24.39\text{kN/m}</math></p> | <p>d= 107mm</p> |

| References                                                                                                       | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Outcomes                                                                                |
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| <p>Table A.8<br/>minimum area of reinforcement</p> <p>BS EN 1992-1:2004<br/>EN 1992-1-1:200(E)<br/>Table:3.1</p> | $\text{Bending Moment}(M_{ed}) = \frac{wl}{8}$ $= \frac{24.39 * 2.93}{8} = 8.93 \text{KNm}$ $\text{Bending Reinforcement (k)} = \frac{M}{bd^2 f_{ck}}$ $= \frac{8.93 \times 10^6}{1000 \times 107^2 \times 37} < 0.167 = 0.0015 \text{ (ok)}$ <p>Determining lever arm,</p> $z = d \left[ 0.5 + \sqrt{0.25 - \frac{k}{1.134}} \right]$ $= d \left( 0.5 + \sqrt{0.25 - \frac{0.0015}{1.134}} \right)$ <p>=0.99d (which is greater than 0.95d) so,</p> $A_s = \frac{M}{0.87 \cdot f_{yk} \cdot 0.95d} = \frac{8.93 \times 10^6}{0.87 \times 500 \times 0.95 \times 107}$ $= 201.95 \text{mm}^2$ <p>Provide H6 bars at 125mm spacing where <math>A_s = 226 \text{mm}^2</math></p> <p>Checking span effective depth ratio</p> $\rho = \frac{100 \cdot A_s}{bd} = \frac{100 \times 202}{1000 \times 107}$ $= 0.18 \% \quad (> 0.15 \% \text{ minimum requirement})$ <p>Minimum and maximum Reinforcement</p> $A_{s \min} = 0.26 \left( \frac{f_{ctm}}{f_{yk}} \right) bd$ $= 0.26 \left( \frac{2.9}{500} \right) 1000 \times 107$ $= 161.356 \text{mm}^2$ $A_{s \max} = 0.04 \cdot A_c$ $\equiv 0.04 \times 1000 \times 140$ $= 5600 \text{mm}^2/\text{m}$ <p>Secondary bar 6mm at 150mm spacing <math>A_{s \min}</math> is <math>162 \text{mm}^2</math></p> <p><b>For SHEAR</b></p> <p>Design shear force <math>V_{ed} = 24.39 \text{KN}</math></p> $V_{Rdc} = [0.12k (100\rho_1 f_{ck})^{\frac{1}{3}}] bd$ <p>Now,</p> $K = (1 + 200/d)^2$ $= (1 + 200/107)^{0.5} = 1.69 \leq 2 \text{ (ok)}$ | <p><math>A_s = 226 \text{mm}^2</math></p> <p><math>V_{Rdc} = 51.03 \text{KN}</math></p> |

| References                                                   | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Outcomes            |
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| BS EN 1992-1-2:2004<br>EN 1992-1-1:2004<br>Table:7.4N pg 128 | $\rho_1 = \frac{A_s}{bd} = \frac{226}{1000 \times 107} \leq 0.02 = 0.002 \text{ (ok)}$ $V_{Rdc} = [0.12k (100\rho_1 f_{ck})^{\frac{1}{3}}]bd$ $= [0.12 \times 1.69 (100 \times 0.002 \times 500)^{1/3}]1000 \times 107$ $= 51.03KN$ $V_{min} = [0.035K^{1.5} \cdot f_{ck}^{0.5}] \cdot bd$ $= [0.0035 \times 1.69^{1.5} \cdot 37^{0.5}]1000 \times 107$ $= 50.04kN$ $V_{Rdc} = 51.03KN > V_{ed} = 24.39KN \text{ (ok)}$ <p>Hence, it passes the shear check.</p> <p><b>Deflection check</b></p> $\rho = \frac{A_{s,req}}{bd} = \frac{201.95}{1000 \times 107} = 0.0018$ <p>Reference reinforcement ratio,</p> $\rho_0 = f_{ck} \times 10^{-3}$ $= 37 \times 10^{-3} = 0.006$ <p>For structural system k=1</p> $(l/d)_{basic} = K \left[ 11 + 1.5\sqrt{f_{ck}} \frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}} \left( \frac{\rho_0}{\rho} \right)^{1.5} \right]$ $= 1[11 + 69.35 + 30.40] = 110.75$ <p>Modification factor = 1.0</p> <p>Modification factor for steel area provided = <math>A_{s,prov}/A_{s,req}</math></p> $= 226/201.95$ $= 1.1 < 1.5 \text{ (ok)}$ <p>Therefore, allowable span-effective depth ratio:</p> $(L/D)_{allowable} = 110.75 \times 1.1 \times 1$ $= 121.85$ <p>Actual span effective depth-ratio:</p> $(L/D)_{actual} = \frac{3000}{107} = 28.38 < (L/D)_{allowable} \text{ (ok)}$ | $V_{min} = 50.04Kn$ |

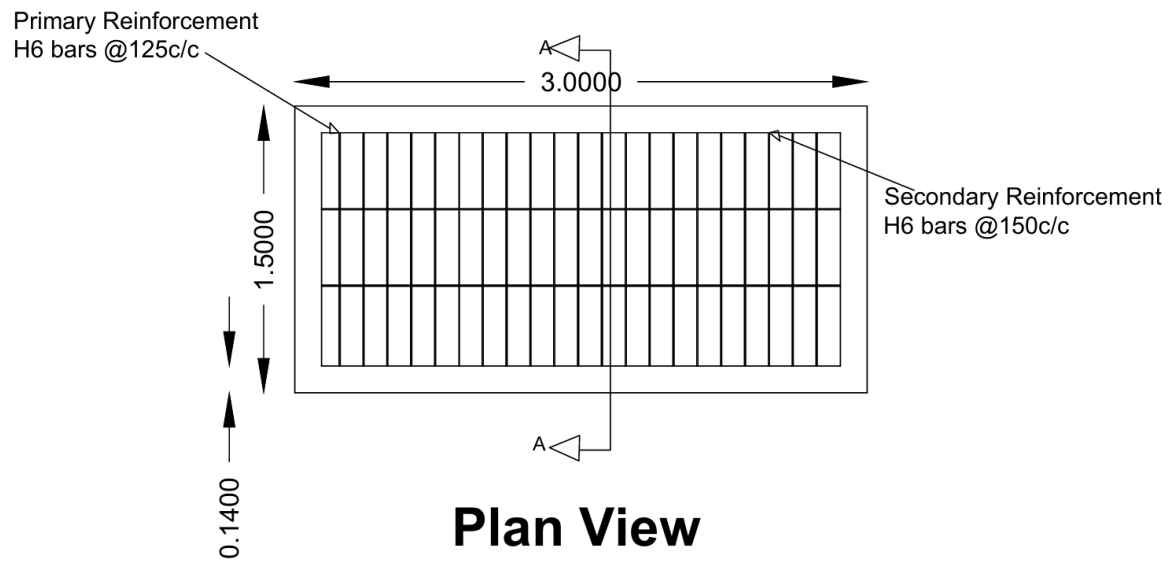


Fig: Plan view of a one-way slab

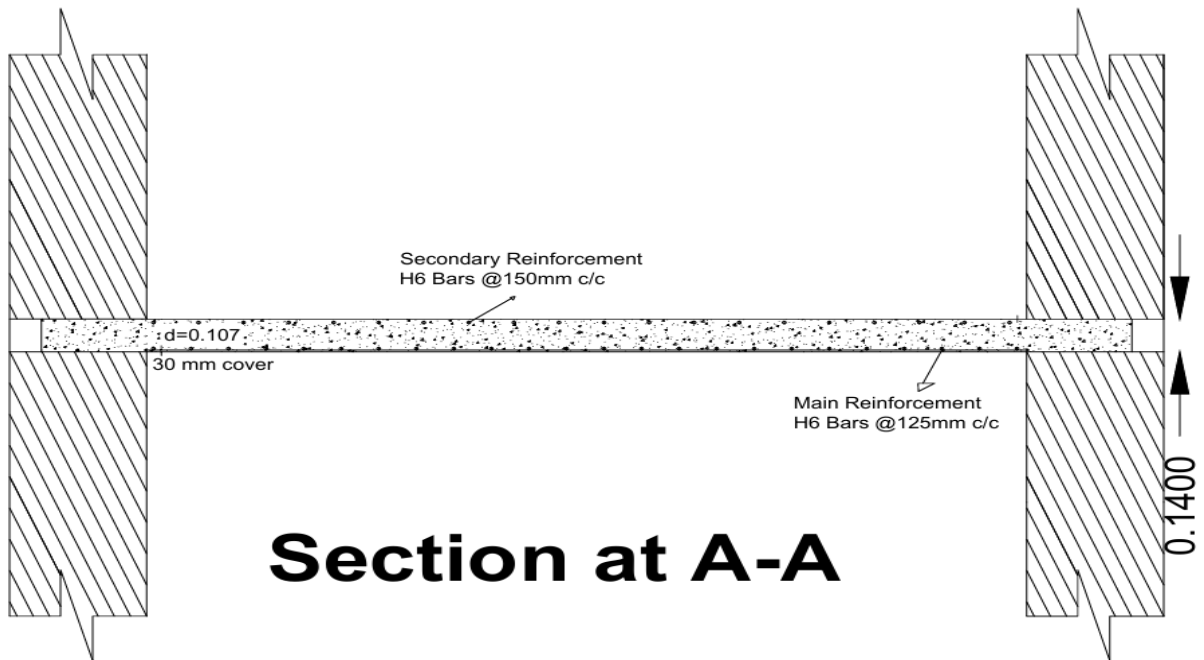


Fig: Section of a slab

| References                                                                                                                                                                                      | Calculation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Outcomes                                                                                                                                                                                                                                                             |
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| <p>BS EN 338:Table2.5</p> <p>BS EN 1991-1-1:2002 EN 1991-1-1:2002 Table A3</p> <p>BS EN 1995-1-1:2004+A1:2008 EN 1995-1-1:2004+A 1:2008</p> <p>Table :3.1</p> <p>Clause3.2</p> <p>Table:2.3</p> | <p><b>A4. Timber Floor Joist Design (EN 1995)</b></p> <p>Density of a materials(<math>\rho</math>) = <math>350\text{kg}/\text{m}^3</math></p> <p>Span of a beam = <math>3\text{m}</math></p> <p>Spacing of a joist = <math>400\text{mm}</math></p> <p>Now,</p> <p>Self-weight of a beam = <math>\rho b h g</math></p> $= 350 \times 0.047 \times 0.15 \times 9.81$ $= 0.024\text{Kn}/\text{m}$ <p>Dead load per joist = <math>0.024 \times 0.4</math></p> $= 0.0096\text{ KN}/\text{m}$ <p>Load for Floor finish</p> $= 0.50 \times 0.4 = 0.2\text{kn}/\text{m}$ <p>Load for Floor decking 18 mm</p> $= 6.5 \times 0.018 \times 0.4 = 0.0468\text{Kn}/\text{m}$ <p>Load for Ceiling (12.5mm plasterboard)</p> $= 0.0125 \times 8.5 \times 0.4 = 0.0425\text{Kn}/\text{m}$ <p>Total dead load = <math>0.024 + 0.0096 + 0.2 + 0.0468 + 0.0425</math></p> $= 0.329\text{Kn}/\text{m}$ <p>Live load = <math>1.5\text{KN}/\text{m}^2</math></p> <p>Live load per joist = <math>1.5 \times 0.4</math></p> $= 0.6\text{Kn}/\text{m}$ <p>Design load = <math>1.35 \times 0.329 + 1.5 \times 0.6</math></p> $= 1.34\text{Kn}/\text{m}$ <p>Design bending stress,</p> $M_d = \frac{wl^2}{8} = \frac{1.34 \times 3^2}{8} = 1.50\text{Knm}$ <p>Section of modulus of beam (y-y)</p> $W_y = \frac{bd^2}{6} = \frac{47 \times 150^2}{6} = 0.176 \times 10^6\text{mm}^3$ <p>Design bending stress</p> $\sigma_{myd} = \frac{M_d}{w_y} = \frac{1.50 \times 10^6}{0.176 \times 10^6} = 8.5\text{N}/\text{mm}^2$ <p>Design bending strength</p> $f_{myd} = \frac{k_{mod} \cdot k_{sys} \cdot k_h \cdot f_{mk}}{\gamma_m}$ <p>Where,</p> <p><math>k_{mod} = 0.8</math> (service class 1)</p> <p><math>k_{sys} = 1</math></p> | <p>Design load<br/><math>1.34\text{Kn}/\text{m}</math></p> <p><math>M_d = 1.50\text{Knm}</math></p> <p><math>W_y</math><br/><math>= 0.176</math><br/><math>\times 10^6\text{mm}^3</math></p> <p><math>f_{myd}</math><br/><math>= 17.6\text{N}/\text{mm}^2</math></p> |

| References                                                                                   | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Outcomes                                                       |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| BS EN 1995-1-1:2004+Al:2008<br>EN 1995-1-1:2004+A1:2008<br>Table :3.1<br>BS EN 338 table 2.5 | $k_h = \left(\frac{600}{h}\right)^{0.1} = \left(\frac{600}{150}\right)^{0.1} = 1.1$ $\gamma_m = 1.2 \text{ (partial factor for LVL material)}$ $= \frac{0.8 \times 1 \times 1.1 \times 24}{1.2} = 17.6 \text{ N/mm}^2$ <p>Unity check</p> $= \frac{\sigma_{myd}}{f_{myd}} = \frac{8.5}{17.6} = 0.4 \leq 1 \text{ (ok)}$ <p>Hence, the bending stress satisfies for the 47× 150mm joist</p> <p><b>Calculation for shear stress</b></p> <p style="text-align: center;">Design Requirement</p> $\tau_d \leq f, v, d$ $\tau_d = \frac{3v}{2b_{ef}d} \quad f, v, d = \frac{k_{mod} \times k_{sys} \times f_{v,k}}{\gamma_m}$ $k_{mod} = 0.8 \text{ (service class 1)}$ $k_{sys} = 1$ $f_{v,k} = 2.5$ $\gamma_m = 1.2 \text{ (partial factor for LVL material)}$ $K_{cr} = 1$ <p>Now,</p> $b_{ef} = K_{cr} \times b = 1 \times 47 = 47 \text{ mm}$ $V = \frac{wl}{2} = \frac{1.34 \times 3}{2} = 2000 \text{ N}$ <p>Again,</p> $\tau_d = \frac{3 \times 2000}{2 \times 47 \times 150} = 0.42 \text{ N/mm}^2$ $f, v, d = \frac{0.8 \times 1 \times 2.5}{1.2} = 1.67 \text{ N/mm}^2$ <p>Here,</p> $\tau_d \leq f, v, d$ <p>As the design shear stress is lower than the design shear strength, the joist is sufficient in terms of shear.</p> | $\tau_d = 0.42 \text{ N/mm}^2$ $f, v, d = 1.67 \text{ N/mm}^2$ |

## Floor Joist Design

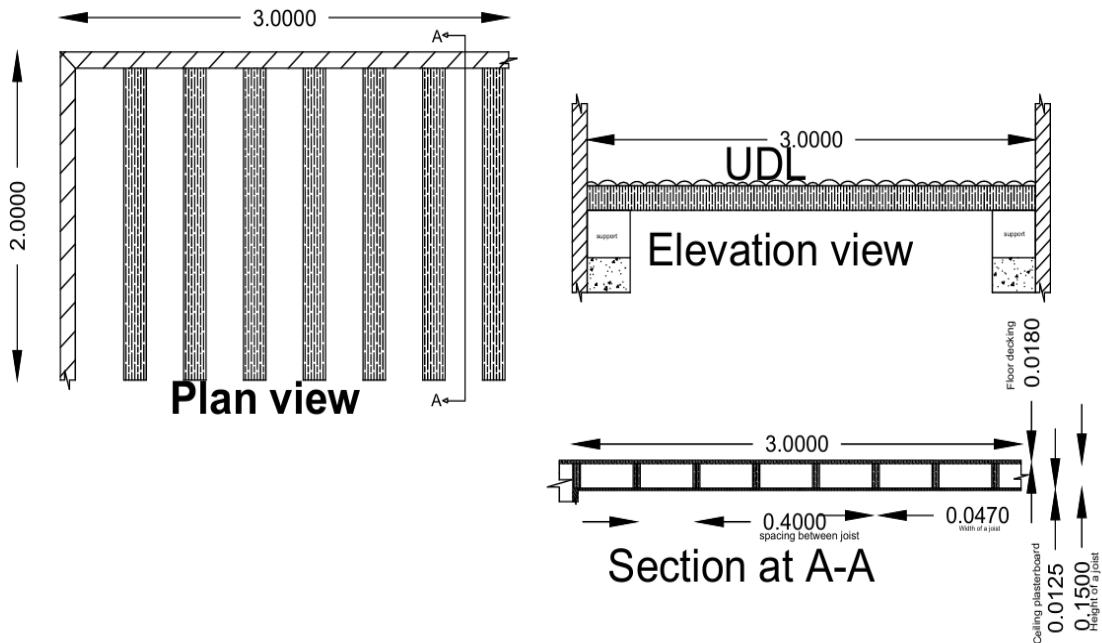


Fig: Technical Drawing for Joist

| References | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | outcomes          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|            | <p style="text-align: center;"><b>A5. Masonry Wall Panel Check</b></p> <p>Ultimate Design load = 31.45Kn/m</p> <p>Length of a wall = 3m</p> <p>Height of a wall = 2.6m</p> <p>Thickness of a wall=100mm</p> <p>Now,</p> $H_{ef} = \rho_n h = 0.75 \times 2600 = 1950mm$ <p>Single leaf actual thickness(<math>T_{ef}</math>) = 100mm</p> <p>Now,</p> $\text{Slenderness Ratio} = \frac{H_{ef}}{T_{ef}} = \frac{1950}{100} = 19.5 < 27$ <p>The effect of creep can be ignored:<br/>i.e. <math>e_k = 0</math></p> <p>Capacity reduction factor</p> <p>Eccentricity at top and bottom of a wall</p> $\frac{M_{id}}{N_{id}} = 0, \text{ since it's axially loaded}$ <p><math>e_{he} = 0</math> (since, no any horizontal load)</p> | $H_{ef} = 1950mm$ |

| References | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outcomes |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|            | $e_{init} = \frac{H_{ef}}{450} = \frac{1950}{450} = 4.3mm$ <p>The eccentricity of the design vertical load at the top and bottom of the wall, <math>e_i</math>, is given by</p> $e_i = \frac{M_i}{N_i} + e_{hi} \pm e_{init}$ $= 0 + 0 + 4.3 \geq 0.05t = 0.05 \times 100 = 5mm$ <p>Hence, <math>e_i = 5mm = 0.05t</math></p> <p>The corresponding capacity reduction factor</p> $\Phi_i = 1 - \frac{2e_i}{t} = 1 - \frac{2 \times 5}{100} = 0.9$ <p>Eccentricity in middle of the wall</p> $\frac{M_{md}}{N_{md}} = 0$ $e_{hm} = 0 \text{ since, no horizontal load}$ $e_{init} = \frac{H_{ef}}{450} = \frac{1950}{450} = 4.3mm$ <p>Eccentricity of the design vertical load in the middle of the wall, <math>e_m</math>, is given by</p> $e_m = \frac{M_{md}}{N_{md}} + e_{hm} \pm e_{init}$ $= 0 + 0 + 4.3 \geq 0.05t = 0.05 \times 100 = 5mm$ <p>Total eccentricity in the middle of the wall</p> $, e_{mk} = e_m + e_k = 5. + 0 = 0.05t$ <p>Modulus of elasticity <math>\epsilon = 500fk</math></p> $u = \frac{\frac{h_{ef}}{t_{ef}} - 2}{23 - 37 \cdot \frac{e_{mk}}{t}} = \frac{19.5 - 2}{23 - 37 \times \left( \frac{0.05 \times 100}{100} \right)} = 0.827$ $A_1 = 1 - 2 \cdot \frac{e_{mk}}{t} = 1 - 2 \times \frac{0.05}{t} = 0.9$ $\varphi_m = A_1 \times e^{-\frac{u^2}{2}}$ $= 0.9 \times e^{-\frac{0.827^2}{2}} = 0.63(ok)$ <p>DESIGN VERTICAL LOAD RESISTANCE OF WALL (<math>N_{Rd}</math>)</p> <p><b>Modification factor</b></p> <p>Small plan area modification factor does not apply since horizontal cross-sectional area of wall, <math>A = 0.1 \times 3.0 = 0.3m^2 &gt; 0.1m^2</math>.</p> <p><b>Safety factor for materials (<math>\gamma_m</math>)</b></p> |          |

|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                              |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <p>BS EN 1996-1-1:2005 EN 1996-1-1:2005 (E)<br/>Table 3.3</p> <p>EN1996-1-1<br/>Clause A.1</p> | <p>Since manufacturing control is Category I and execution control is Class 1, from Table 10.6 <math>\gamma_m = 2.3</math></p> <p><b>Design vertical load resistance</b></p> $N_{RD} = \frac{\varphi_{i,m} \cdot t \cdot f_k}{\gamma_m}$ $= \frac{0.63 \times 100 \times f_k}{2.3}$ <p>Determining <math>f_k</math>,<br/>For structural stability</p> $\frac{N_{Rd} \geq N_{Ed}}{= \frac{0.63 \times 100 \times f_k}{2.3} \geq 31.45}$ $f_k \geq \frac{31.45}{27.39} = 1.14 \text{ N/mm}^2$ <p><b>SELECTION OF BLOCK AND MORTAR</b><br/>TYPE Assuming the wall is made of category 1 of standard format size laid in M4 general purpose mortar<br/> <math>\Rightarrow K = 0.55</math> (Aggregate concrete group 1)<br/> <math>\Rightarrow f_m = 6 \text{ N/mm}^2</math><br/> <math>\Rightarrow</math> shape factor = 0.85<br/> <math>\Rightarrow</math> conditioning factor = 1 (assuming units are manufactured to EN 771-1)<br/> The characteristic strength of masonry, <math>f_k</math>, is given by</p> $f_k = K \cdot f_b^{0.7} \cdot f_m^{0.3}$ <p>Or, <math>1.14 = 0.55 \times f_b^{0.7} \times 6^{0.3} = 1.216 \text{ N/mm}^2</math><br/> Therefore, the normalized compressive strength of the units, <math>f_b \geq 1.21 \text{ N/mm}^2</math><br/> The declared mean compressive strength of the units is given by:<br/> <math>f_b = \text{conditioning factor} \times \text{shape factor} \times \text{declared mean compressive strength}</math><br/> or, <math>1.2 = 1 \times 0.85 \times \text{declared mean compressive strength}</math></p> <p>Therefore, the required declared dried mean compressive strength of units <math>\geq 1.42 \text{ N/mm}^2</math></p> | <p><math>f_k = 1.14 \text{ N/mm}^2</math></p> <p><math>f_b = 1.216 \text{ N/mm}^2</math></p> |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|

## Masonry Wall Design

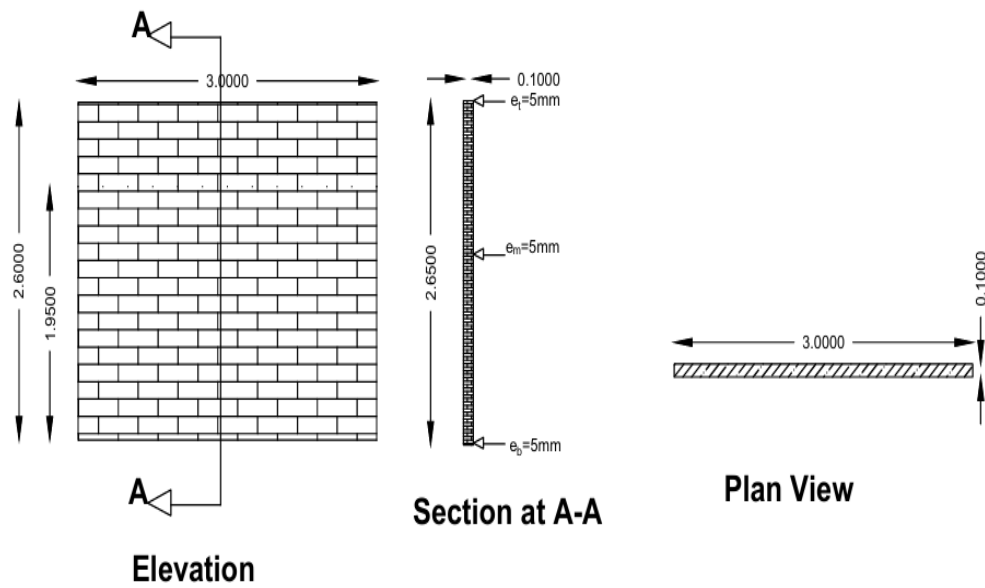
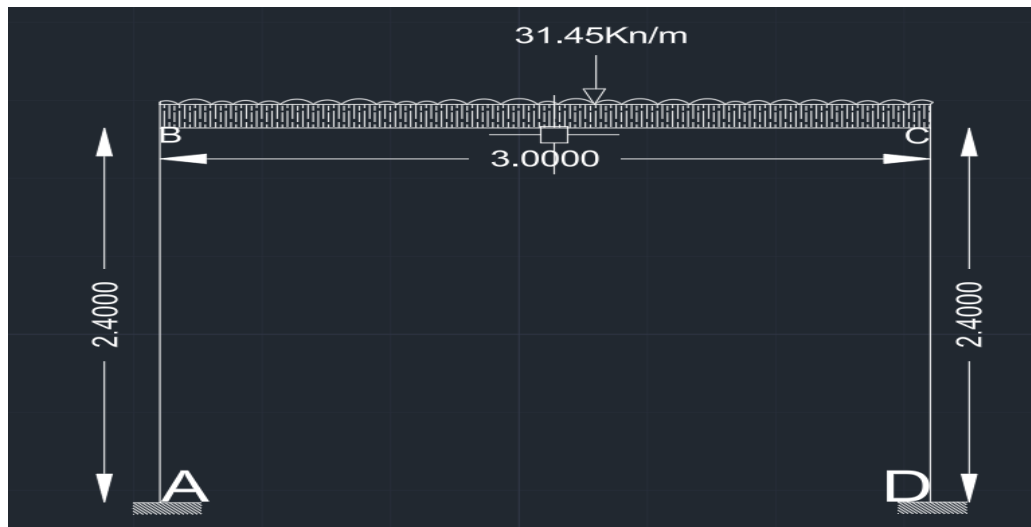


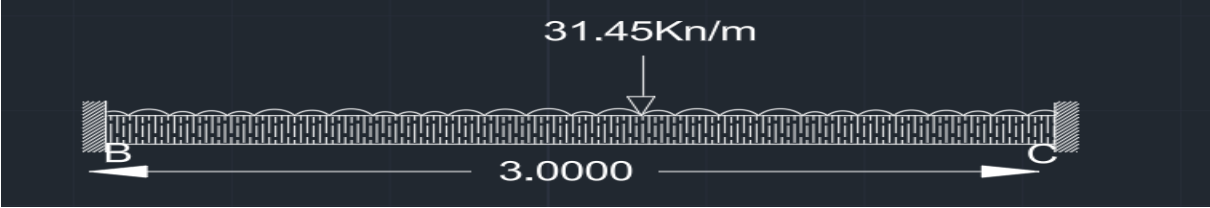
Fig: Technical Drawing for Masonry Wall

## SECTION B – FRAME ANALYSIS: MOMENT DISTRIBUTION



| References | Calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Outcomes |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|            | <p style="text-align: center;"><b>Stiffness Factor calculation</b></p> $K_{BA} = \frac{4EI}{L} = \frac{4EI}{2.4}$ $K_{BC} = \frac{4EI}{L} = \frac{4EI}{3}$ $K_{CB} = \frac{4EI}{4} = \frac{4EI}{3}$ $K_{CD} = \frac{3EI}{L} = \frac{4EI}{2.4}$ <p style="text-align: center;"><b>Distribution Factor Calculations</b></p> $DF_{BA} = \frac{\frac{4EI}{2.4}}{\frac{4EI}{2.4} + \frac{4EI}{3}} = 0.5$ $DF_{BC} = \frac{\frac{4EI}{3}}{\frac{4EI}{2.4} + \frac{4EI}{3}} = 0.4$ $DF_{CB} = \frac{\frac{4EI}{3}}{\frac{4EI}{3} + \frac{3EI}{2.4}} = 0.4$ $DF_{CD} = \frac{\frac{4EI}{2.4}}{\frac{4EI}{3} + \frac{4EI}{2.4}} = 0.5$ |          |

**Fixed End Moments Calculation**



$$Fem_{BC} = \frac{-wl^2}{12}$$

$$= \frac{-31.45 \times 3^2}{12}$$

$$Fem_{CB} = \frac{wl^2}{12}$$

$$= \frac{31.45 \times 3^2}{12}$$

$= 23.58 \text{ kNm}$

| Moment Distribution Calculation |          |            |              |             |             |           |
|---------------------------------|----------|------------|--------------|-------------|-------------|-----------|
| joint                           | A        | B          |              | C           |             | D         |
| Member                          | AB       | BA         | BC           | CB          | CD          | DC        |
| DF                              | 0        | 0.5        | 0.4          | 0.4         | 0.5         | 0         |
| FEM                             |          |            | -23.58       | 23.58       |             |           |
| Dist                            |          | 11.79      | 9.432        | -9.432      | -11.79      |           |
| Co                              | 5.895    |            | -4.716       | 4.716       |             | -5.895    |
| Dist                            |          | 2.358      | 1.8864       | -1.8864     | -2.358      |           |
| Co                              | 1.179    |            | -0.9432      | 0.9432      |             | -1.179    |
| Dist                            |          | 0.4716     | 0.37728      | -0.37728    | -0.4716     |           |
| Co                              | 0.2358   |            | -0.18864     | 0.18864     |             | -0.2358   |
| Dist                            |          | 0.09432    | 0.075456     | -0.075456   | -0.09432    |           |
| Co                              | 0.04716  |            | -0.037728    | 0.037728    |             | -0.04716  |
| Dist                            |          | 0.018864   | 0.0150912    | -0.0150912  | -0.018864   |           |
| Co                              | 0.009432 |            | -0.0075456   | 0.0075456   |             | -0.009432 |
| Dist                            |          | 0.0037728  | 0.00301824   | -0.00301824 | -0.0037728  |           |
| Total                           | 7.366392 | 14.7365568 | -17.68386816 | 17.68386816 | -14.7365568 | -7.366392 |

Fig: Moment Distribution Table

## SECTION C – SPACE GASS FRAME ANALYSIS

### C1. Loading Condition

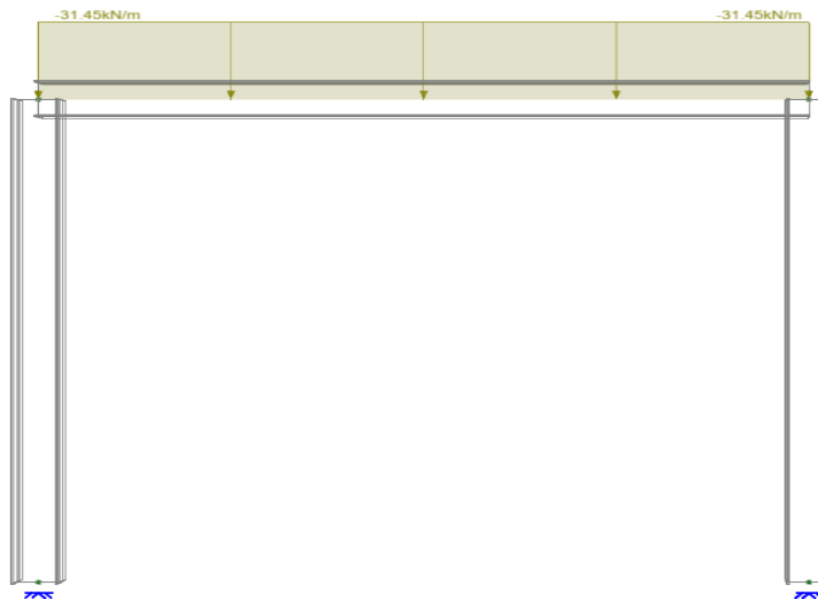


Fig: loading condition for frame

## C2. Bending Moment Diagram

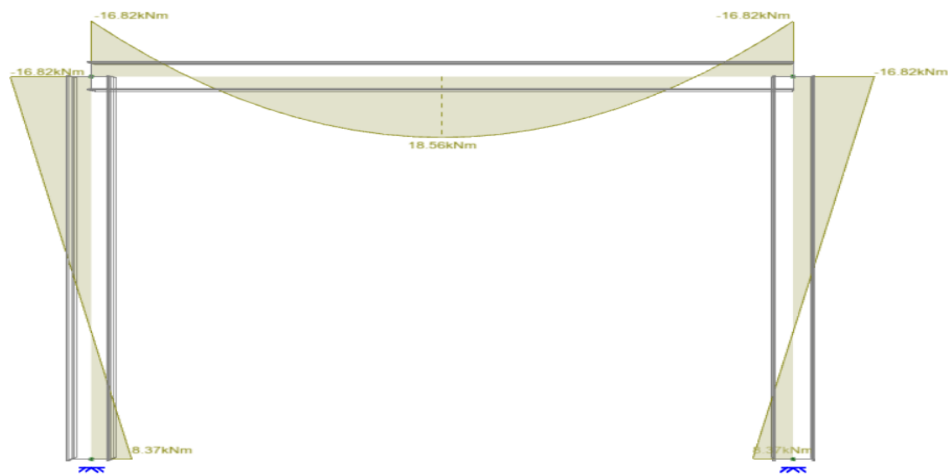


Fig: Bending Moment

## C3. Shear Force Diagram

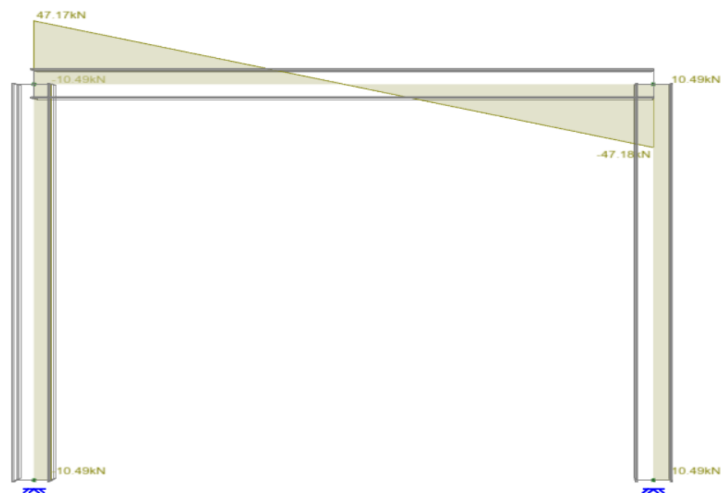


Fig: Shear Force Diagram

**References for A1**

McCormac, J.C. & Brown, R.H. – *Design of Reinforced Concrete*

Chudley & Greeno – *Building Construction Handbook*